Training-Free Persona-Based Korean Voter Simulation under Candidate-Information Priors

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Abstract

Large language models are increasingly applied to voter simulation, but how candidate names, policy descriptions, and party labels shape persona-following behavior remains poorly understood. We study training-free persona-based simulation of the 2025 Korean presidential election under a name-exposed ballot. Through controlled anonymization experiments that progressively remove candidate names, policy descriptions, and party labels, we find that simulated vote-share distributions shift substantially as these cues are withheld, especially through reduced overprediction of the progressive candidate. We interpret this not as evidence that anonymized ballots provide a faithful election simulation, but as a diagnostic indication that candidate and party-name priors, together with cues about ideological direction, strongly shape model outputs in name-exposed settings. To address this within a realistic simulation, we propose two training-free interventions: Belief Network Seeding (BNS), which augments voter profiles with KGSS-derived issue-level beliefs, and Persona Vector (PV) steering, which reinforces assigned political orientation at the activation level. BNS most clearly affects Moderate agents in the name-exposed setting, reducing their Lee Jae-myung support from 86.0% to 76.5% and thereby mitigating progressive collapse. PV alone provides smaller and inconsistent gains. The combined BNS+PV condition achieves the best national, sex, and age MAE in the name-exposed setting, while BNS alone performs slightly better at the regional level. Diagnostic anonymization results further show that BNS remains the more reliable component, whereas PV alone rarely improves over the base condition.

1 Introduction

The use of large language models as synthetic survey respondents has attracted sustained attention since Argyle et al. showed that demographic conditioning can reproduce human response distributions at scale [1]. For political science, the appeal is practical: scalable voter simulation without repeatedly fielding large surveys. However, simulation quality is not uniform: Qu and Wang find that accuracy degrades for non-Western electorates and increases sharply with the number of response options [2], while Li et al. argue that aggregate accuracy often masks structural failures within demographic subgroups [3]. In Korean voter simulation specifically, Kang et al. identify progressive bias among moderate agents, third-party candidate collapse, and regional polarization collapse across six certified elections [4]. These findings suggest that careful demographic grounding is necessary but not sufficient.

We study a version of this problem that arises naturally in realistic settings: what happens when real candidate names and policy information appear in the prompt? An LLM's representation of a well-known political figure is not neutral—training corpora encode associations between specific candidates, parties, and ideological positions that can be stronger than the context provided at inference time. Prior work also reports progressive-leaning tendencies in frontier LLMs [5, 6], which may amplify candidate-level associations toward well-known progressive figures in name-exposed election settings. When these associations conflict with the persona assigned to a synthetic voter, the model may bypass the persona and default to its own candidate-level priors. Kreutner et al. document a closely related problem in European Parliament simulation, where prediction quality drops for political groups whose assigned personas diverge from the model's default tendencies [7].

To understand how different types of candidate information affect simulated vote distributions, we run controlled anonymization experiments in which candidate names, policy descriptions, and party labels are progressively removed from the prompt. These reduced-information settings are not deployable simulations; they serve as diagnostic controls for measuring sensitivity to candidate-related cues. As candidate information is withheld, simulated vote shares move away from the progressive-candidate overprediction observed in the fully name-exposed setting, suggesting that candidate and party-name priors, together with cues about ideological direction, strongly shape model outputs. However, anonymization is not a viable solution: even under complete anonymization, agents hallucinate policy positions consistent with training-data knowledge of Korean parties, partially reconstructing candidate identities from context alone.

We therefore propose two training-free persona-preservation methods for the realistic name-exposed setting. Belief Network Seeding (BNS) augments each voter agent's profile with KGSS-derived issue-level belief statements, grounding the assigned orientation in concrete attitudes rather than an abstract label. Persona Vector (PV) steering, adapted from Chen et al. [8], injects orientation-specific activation directions at inference time as an activation-level complement to BNS. We evaluate both methods through a five-condition ablation study against official 2025 national, regional, and demographic subgroup results.

Our contributions are threefold. First, we build a training-free Korean voter simulation framework using Nemotron-Personas-Korea narratives and KGSS-conditional political orientations under a name-exposed 2025 election setting. Second, we use controlled anonymization to analyze how candidate names, policy descriptions, and party labels reshape simulated vote distributions, showing why anonymization cannot substitute for realistic name-exposed simulation. Third, we find that BNS substantially reduces progressive collapse among moderate agents, while PV provides complementary gains mainly in the combined condition.

2 Related Work

LLMs as silicon samples. Argyle et al. introduced silicon sampling, showing that language models conditioned on demographic backstories can reproduce known survey response distributions across multiple political dimensions [1]. Subsequent work has shown that this capability is not uniformly reliable. Qu and Wang report that simulation accuracy degrades for non-Western countries and becomes harder as the number of response options increases [2], while Li et al. argue that aggregate accuracy can hide failures in subgroup structure and within-group diversity [3]. These critiques motivate our multi-level evaluation beyond national vote-share accuracy and our use of direct KGSS respondent resampling to preserve within-group heterogeneity.

Korean voter simulation. Kang et al. provide the most direct prior work on Korean LLM-based voter simulation, constructing census-grounded Korean voter agents and evaluating LLMs across certified elections [4]. They identify systematic failures such as progressive bias among moderate agents, third-party candidate collapse, and regional polarization collapse. Their mitigation ultimately relies on supervised output-side calibration using labeled election outcomes. We share their demographic grounding and subgroup evaluation design, but focus on whether training-free persona-preservation methods can reduce distortion under candidate-identifying prompts.

Persona fragility in political simulation. Persona-driven simulation assumes that the model follows the assigned persona rather than its default political tendencies. Kreutner et al. show that voting simulation quality varies across political groups, with models performing better when assigned personas align with their default tendencies and worse when they diverge [7]. This is closely related to our candidate-information prior problem: real candidate names, parties, and policies can activate pretrained associations that overpower the assigned voter persona. Our analysis therefore focuses on when persona signals are preserved or suppressed under different levels of candidate-identifying information.

Inference-time persona steering. Prompt-based persona conditioning can be fragile when pretrained priors are strong. Chen et al. propose Persona Vectors, which identify activation-space directions associated with character traits and inject them during inference [8]. We adapt this idea from general character-trait control to Korean political-orientation preservation by constructing conservative-progressive steering directions. In our framework, PV thus serves as an activation-level complement to BNS's prompt-level belief grounding.

3 Approach

3.1 Task Formulation

We formulate Korean voter simulation as a conditional vote-generation task. Each synthetic voter agent is assigned demographic attributes, a narrative persona, and a political orientation. Given a name-exposed ballot and candidate information for the 2025 Korean presidential election, the model generates a vote choice for each agent. The evaluation target is distributional: we assess whether generated vote shares match official results at national, regional, and demographic subgroup levels.

3.2 Demographic-Conditional Agent Construction

Narrative profiles are sampled from Nemotron-Personas-Korea [9], and each agent is assigned age, sex, region, education, and a five-way political orientation: Very Progressive (VP), Progressive (P), Moderate (M), Conservative (C), or Very Conservative (VC). Following Kang et al. [4], orientations are sampled from demographic-conditional KGSS distributions with coarser fallback cells for sparse groups.

To reduce conflicts between narratives and assigned orientations, an auxiliary LLM judge assigns each narrative a six-way political lean, (VP/P/M/C/VC/AMBIGUOUS) with a confidence score in $\{1, \ldots, 5\}$; agent pairs with orientation–lean distance ≥ 2 and confidence ≥ 4 are eligible for swap within the same (sex, age bracket, sido) cell, preserving the joint demographic–orientation distribution.

3.3 Belief Network Seeding

Orientation labels provide only a coarse specification of voter behavior. Two voters labeled as conservative may differ substantially on redistribution, national security, or market competition, and these differences can matter in a three-candidate race. BNS addresses this by augmenting each agent's profile with short issue-level belief statements derived from KGSS [10].

We first use political attitude items from the 2016 KGSS module to estimate latent belief dimensions (economic redistribution, North Korea and security, market liberalization, political efficacy). For each agent, we directly resample a real KGSS respondent from the corresponding demographic-orientation cell and assign that respondent's factor scores. Direct resampling is preferred over Gaussian sampling in order to preserve real within-group heterogeneity [3]. The strongest factors are then converted into short natural-language statements and appended to the agent profile.

Condition	Added components	Inference
C_{narr}	Narrative persona only; no explicit political orientation	Standard vLLM
C_{base}	Political-orientation block and bias-mitigation instructions	Standard vLLM
C_{bns}	C_{base} + BNS belief statements	Standard vLLM
C_{pv}	C_{base} + PV steering	vLLM with PV hook
C_{full}	C_{base} + BNS + PV steering	vLLM with PV hook

Table 1: Five-condition ablation design. C_{narr} tests whether narrative personas alone suffice; C_{base} is the main prompting baseline; C_{bns} and C_{pv} isolate the two interventions; C_{full} tests their combination.

3.4 Persona Vector Steering

BNS strengthens the persona signal at the prompt level, but the model may still rely on pretrained candidate associations during generation. Persona Vector (PV) steering [8] provides a complementary intervention at the activation level.

We construct conservative and progressive steering directions from contrastive prompts. For each validated prompt-response pair, we extract the response-token-averaged hidden activation at layer ℓ. Let μ_C^ℓ and μ_P^ℓ denote the mean activations over conservative and progressive sets; we define

$$v_C^\ell = \mu_C^\ell - \mu_P^\ell, \quad v_P^\ell = -v_C^\ell$$

Layer and steering strength are selected using separation and causal-effect metrics before main inference. During generation, the hidden state for an agent with assigned orientation o_i is modified as

$$h'_{\ell,t} = h_{\ell,t} + \alpha_i v_{o_i}^\ell$$

where α_i scales with orientation strength (VP/VC receive higher coefficients than P/C; M agents receive no injection).

3.5 Ablation Conditions

Table 1 summarizes the five conditions, all run on the same synthetic voter population and the same name-exposed ballot.

4 Experiments

4.1 Data

We use four data sources. Nemotron-Personas-Korea provides Korean narrative personas [9]. KGSS provides demographic-orientation distributions and political attitude items for BNS factor estimation [10]. The 2025 election ballot names three candidates—Lee Jae-myung, Kim Moon-soo, and Lee Jun-seok—with the same candidate information used in all name-exposed conditions. National and regional ground-truth vote shares are taken from official NEC results [11]; demographic subgroup shares (by sex and age group) are taken from the KEP joint exit poll [12]. Since our simulation covers only the three major candidates, all ground-truth vote shares are renormalized to the three-candidate total before computing MAE; the remaining $\sim$1–1.6% cast for minor candidates is excluded. After renormalization, the national target is 49.96% / 41.60% / 8.43% for Lee Jae-myung, Kim Moon-soo, and Lee Jun-seok respectively. We note that subgroup-level targets (by sex and age) come from exit-poll estimates rather than certified counts, and are therefore subject to sampling error; MAE figures at the subgroup level should be interpreted accordingly.

4.2 Evaluation Method

For groups $\mathcal{G}$ and candidates $\mathcal{C}$, vote-share mean absolute error (MAE) under condition m is

$$\mathrm{MAE}^{(m)}(\mathcal{G}) = \frac{1}{|\mathcal{G}||\mathcal{C}|} \sum_{g\in\mathcal{G}} \sum_{c\in\mathcal{C}} \left|\hat{p}_{g,c}^{(m)} - p_{g,c}\right|. \tag{1}$$

We report national MAE against the certified NEC result, regional MAE averaged naively over 17 *sido*, sex MAE averaged naively over male and female groups, and age MAE averaged naively over age-group cells from exit-poll data. Condition-level estimates aggregate $K = 5$ samples per agent, and bootstrap confidence intervals are computed by resampling agent-level outputs.

4.3 Experimental Details

All main vote-generation experiments use Qwen3-30B-A3B [13] without any fine-tuning. We run all five conditions on 5,000 synthetic voter agents, generating $K = 5$ vote samples per agent per condition (125,000 total). Narrative-only, base, and BNS conditions use standard vLLM inference [14]; PV and full conditions use a modified backend with a forward-hook injector. Temperature is set to 0.5; maximum generation length is 200 tokens for C_{narr} and 400 tokens for the other conditions. PV contrastive prompts are validated by EXAONE-4.0-32B as a cross-model judge before activation extraction, and layer selection and steering strength are fixed prior to main inference.

To analyze the role of candidate-information priors, we additionally run a *candidate-information prior experiment* on a stratified subsample of 500 agents across three information levels. L_1 removes real candidate names but retains policy descriptions and party labels. L_2 further removes policy descriptions, retaining only party labels. L_3 removes all candidate-identifying information.

4.4 Results

Orientation-level vote distributions. Table 2 reports vote-share distributions broken down by political orientation. C_{narr} collapses to Lee Jae-myung at or above 99.7% across every orientation group, confirming that narrative personas alone are insufficient to preserve ideological differentiation in the name-exposed setting. C_{base} recovers the gross conservative-progressive split—C and VC agents shift toward Kim Moon-soo at 83.5% and 97.0%—but Moderate agents remain skewed toward Lee at 86.0%. BNS has its clearest effect here: C_{bns} brings Moderate support for Lee down to 76.5%, with Kim rising to 20.6%, and this shift is preserved in C_{full} (75.6% / 21.6%). Since Moderate agents receive no direct PV injection by design (Section 3.4), the near-identical distributions between C_{base} and C_{pv} among Moderate agents are expected and do not themselves constitute evidence that activation steering fails without prompt-level support.

National, regional, sex, and age results. Table 3 reports MAE across four levels of aggregation. Subgroup-level targets (sex and age) are derived from exit-poll estimates rather than certified counts and are therefore subject to sampling error; the figures should be read as approximate comparisons rather than exact benchmarks.

At the national level, C_{base} brings MAE to 12.26pp from the near-total collapse under C_{narr} (33.32pp). BNS provides the clearest improvement (9.43pp), with C_{full} adding a further gain (9.26pp); the 0.17pp gap between C_{bns} and C_{full} falls within bootstrap confidence interval overlap and should be interpreted cautiously. C_{pv} alone offers little over C_{base} (11.87pp) and provides no improvement at the regional level (13.00pp vs. 12.94pp). At the sex level, C_{bns} and C_{full} improve over C_{base} in aggregate sex MAE (7.75pp and 7.58pp vs. 10.56pp). Age-group MAE follows a similar pattern, with C_{bns} (11.14pp) and C_{full} (10.43pp) improving over C_{base} (12.20pp). The gap between C_{full} and C_{bns} is larger here (0.71pp) than at the national level (0.17pp), suggesting that PV's activation-level contribution, while secondary to BNS, has a slightly more consistent effect when evaluated at the age-group rather than the aggregate level. Regional MAE shows the most modest gains: C_{bns} (11.18pp) and C_{full} (11.41pp) improve over C_{base} (12.94pp), but the gap between the two interventions closes and even reverses.

Candidate-information prior results. Table 4 reports how simulated vote distributions shift as candidate-related information is progressively removed. MAE falls steadily from L_1 through L_3 across all conditions, indicating that simulated vote distributions are highly sensitive to candidate names, party labels, and cues about ideological direction. For C_{full}, MAE drops from 4.96pp (L_1) to 3.60pp (L_2) and 0.93pp (L_3). This pattern should not be read as evidence that anonymized prompts provide a more faithful election simulation; rather, it shows how strongly candidate-related cues can reshape model outputs in this setting.

The type of information removed matters. Removing candidate names (L_1) pulls the progressive candidate's share back toward the official split, while removing policy descriptions (L_2) further

Table 2: Vote-share distributions by political orientation (5,000 agents, name-exposed setting). Lee = Lee Jae-myung, Kim = Kim Moon-soo, Lee JS = Lee Jun-seok. Agg. L/K/J reports the condition-level aggregate vote distribution after weighting orientation groups. Moderate (M) agents are the primary site of intervention effects.

CONDITION	ORIENT	n	LEE (%)	KIM (%)	LEE JS (%)	AGG. L/K/J (%)
C_{NARR}	VP	224	100.0	0.0	0.0	99.9 / 0.1 / 0.0
	P	1076	100.0	0.0	0.0	
	M	2400	100.0	0.0	0.0	
	C	1000	99.7	0.3	0.0	
	VC	300	100.0	0.0	0.0	
C_{BASE}	VP	224	100.0	0.0	0.0	68.4 / 28.1 / 3.5
	P	1076	99.3	0.4	0.4	
	M	2400	86.0	11.5	2.5	
	C	1000	6.1	83.5	10.4	
	VC	300	0.3	97.0	2.7	
C_{BNS}	VP	224	97.8	2.2	0.0	64.1 / 33.1 / 2.8
	P	1076	97.3	2.3	0.4	
	M	2400	76.5	20.6	2.8	
	C	1000	10.1	83.3	6.6	
	VC	300	0.3	99.3	0.3	
C_{PV}	VP	224	100.0	0.0	0.0	67.8 / 28.6 / 3.6
	P	1076	99.3	0.4	0.4	
	M	2400	84.6	12.9	2.5	
	C	1000	6.6	82.7	10.7	
	VC	300	0.0	97.0	3.0	
C_{FULL}	VP	224	98.2	1.3	0.4	63.8 / 33.2 / 2.9
	P	1076	98.0	1.4	0.5	
	M	2400	75.6	21.6	2.6	
	C	1000	9.9	83.2	6.9	
	VC	300	0.3	98.0	1.7	

Table 3: MAE (%p) at four levels of aggregation. All ground-truth shares are renormalized to the three-candidate total before computing MAE. Regional and sex MAE are naive averages over *sido* cells and sex cells respectively; age MAE is a naive average over age-group cells from exit-poll data. Sex and age targets are exit-poll estimates and subject to sampling error. Lower is better.

CONDITION	NATIONAL ↓	REGIONAL ↓	SEX ↓	AGE ↓
C_{NARR}	33.32	32.00	31.70	31.32
C_{BASE}	12.26	12.94	10.56	12.20
C_{BNS}	9.43	**11.18**	7.75	11.14
C_{PV}	11.87	13.00	10.16	11.83
C_{FULL}	**9.26**	11.41	**7.58**	**10.43**

narrows the Lee/Kim gap, suggesting that policy wording itself may carry ideological-direction cues. The largest single gain comes from removing party labels (L_3), which primarily recovers Lee Jun-seok's share; party-label associations appear more closely linked to third-candidate suppression than name or policy cues.

The low MAE at L_3 does not make anonymization a viable simulation strategy. Even with all labels removed, agents can reconstruct candidate identities from training data: one L_3 agent reasoned that "Candidate 1 opposes privatization of public enterprises"—a position absent from the prompt—drawing on internalized knowledge of Korean party platforms. A fully anonymous setup also carries little election-specific signal, so the same configuration could produce similar output for any election with a comparable ideological structure. L_1–L_3 therefore serve as diagnostic controls for measuring sensitivity to candidate-related information, not as deployable simulation methods.

Table 4: National vote-share distributions and MAE (%p) across three information levels (500 stratified agents). L/K/J denotes Lee Jae-myung, Kim Moon-soo, and Lee Jun-seok. L_1 removes candidate names; L_2 additionally removes policy descriptions; L_3 removes all candidate-identifying information. NEC shares renormalized to the three major candidates. Bold marks the lowest MAE per level, excluding C_{narr}.

	L_1: Name Removed		L_2: Name+Policy Removed		L_3: Full Anon	
Condition	L/K/J (%)	MAE	L/K/J (%)	MAE	L/K/J (%)	MAE
C_{NARR}	99.0/0.8/0.2	32.69	84.4/7.6/8.0	22.96	51.4/15.6/33.0	17.34
C_{BASE}	58.0/41.2/0.8	5.36	53.2/44.6/2.2	4.16	51.6/43.8/4.6	2.56
C_{BNS}	54.8/44.8/0.4	5.36	50.0/46.4/3.6	**3.22**	48.4/44.8/6.8	2.13
C_{PV}	58.4/41.2/0.4	5.62	53.2/44.8/1.8	4.36	49.4/45.6/5.0	2.66
C_{FULL}	52.8/46.2/1.0	**4.96**	49.6/47.0/3.4	3.60	49.6/43.0/7.4	**0.93**
NEC actual	49.96 / 41.60 / 8.43					

5 Analysis

To better understand how the interventions affect individual-level reasoning, we inspect two representative Moderate agents.

Agent A (female, 19–29, South Chungcheong, product designer) has KGSS factors leaning progressive on security (-1.328) and market liberalization (-0.517), with BNS seeds emphasizing reform of authoritarian security laws and public enterprise accountability. Across C_{base} and C_{bns}, the agent selects Lee Jae-myung, citing social safety nets and public-sector policies as aligned with her occupational context. The notable departure is C_{pv}, where the agent switches to Lee Jun-seok on grounds of expanded local autonomy and deregulation, a rationale that contradicts her BNS-grounded belief profile. C_{full} returns to Lee, suggesting that prompt-level grounding suppresses this activation-level drift.

Agent B (female, 60–69, South Jeolla, unemployed) has conservative-leaning KGSS factors on redistribution ($+1.093$) and security ($+0.129$), with BNS seeds encoding market autonomy and support for the U.S.–Korea alliance. Despite this profile, C_{bns} selects Lee Jae-myung, reasoning that small-business support, labor rights, and public healthcare best fit the agent's life context. This does not necessarily imply an incorrect prediction; rather, it shows that BNS seeds may be interpreted through regional, demographic, and life-contextual backgrounds instead of mechanically determining vote choice. Here, conservative-leaning issue beliefs are reframed through a welfare and livelihood lens, while C_{base} is the only condition where the agent departs from Lee.

These cases illustrate two different intervention effects. Agent A suggests that PV without BNS grounding can produce reasoning that departs from the assigned belief profile, while Agent B shows that BNS does not force issue-level beliefs into a fixed candidate choice; its effect can be mediated by regional, demographic, and life-contextual cues.

6 Conclusion

This paper examined training-free persona-based simulation of the 2025 Korean presidential election under a name-exposed ballot. Controlled anonymization experiments show that simulated vote distributions are highly sensitive to candidate names, policy descriptions, and party labels. Rather than treating anonymization as a faithful simulation setting, we interpret these reduced-information conditions as diagnostic controls: candidate and party-name priors, together with cues about ideological direction, can strongly reshape model outputs in name-exposed prompts.

Within the realistic name-exposed setting, BNS reduces national MAE from 12.26pp to 9.43pp. The gain is concentrated among Moderate agents, whose Lee Jae-myung support falls from 86.0% to 76.5%, indicating that BNS mitigates progressive collapse by grounding abstract political labels in issue-level beliefs. PV shows limited standalone benefit. Its gains are secondary to BNS and appear mainly in the combined condition, so we treat activation-level steering as a complementary component rather than a reliable standalone correction. Across sex and age subgroups, the same broad ordering holds, with C_{full} and C_{bns} outperforming C_{base}. Regional results are more mixed: C_{bns} achieves the lowest regional MAE (11.18pp), while C_{full} slightly underperforms it (11.41pp), suggesting that PV does not consistently reduce geographically structured errors.

Several limitations constrain what can be drawn from these results. Lee Jun-seok's 8.43% actual share remains substantially under-recovered across all name-exposed conditions, with aggregate support staying below 4% even after BNS or PV interventions. This points to a third-candidate collapse problem that neither prompt-level nor activation-level persona preservation fully resolves. BNS relies on 2016 KGSS data that may not fully capture issue-space shifts in the decade before the 2025 election. All experiments use a single model and a single election, leaving open how well these patterns generalize. Future work should therefore evaluate multiple elections and models while reducing third-candidate suppression and regional errors without removing the candidate-identifying context required for realistic simulation.

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Appendix: Team Contributions

Donghyuk Kang: Contributed to research goal formulation, methodology design, and literature review. Wrote the paper.

Jaemin Kim: Contributed to research goal formulation, methodology design, and literature review. Produced presentation materials.

Hyeonwoo Ma: Contributed to research goal formulation, methodology design, and literature review. Conducted experiments and delivered the final presentation.

Seongeun Lee: Contributed to research goal formulation, methodology design, and literature review. Conducted experiments and wrote the paper.